AP-E10 Exact Stencil No-Go and Finite-R Cell Formula: Discrete Skyrme Defects, Periodic Homogenization, and a Translation-Quotient Continuum Audit

Route F research note

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Abstract

This note closes two AP-E9 questions and gives a negative answer to the first physical conjunction. The one-corner forward Skyrme stencil does not satisfy the compensated-compactness property required by the proposed continuum argument. An exact three-periodic sequence $n_a : \mathbb{T}^3 \to \mathbb{S}^3$ converges uniformly to the vacuum while one of its forward two-minors remains the nonzero constant $3\sqrt{3}\eta^2/2$. A second two-periodic sequence has geometric degree zero but forward topological current converging to $-16\eta^3$. Localizing it with an exact vacuum boundary produces the diffuse defect $-16\eta^3\chi^3\mathrm{d}x$. On the other hand the Skyrme bound implies $|J_a|^{4/3} \leq | \wedge^2 D^a n_a|^2/3$, so singular concentration of this forward current is excluded. The obstruction is a diffuse lattice-scale aliasing defect, not loss of mass into an atomic measure.

The finite-R barrier problem is solved exactly. On both the uniform six-tetrahedron graph and the two checkerboard five-tetrahedron phases, every bond-direction subgraph is cycle-balanced. Direction-wise Jensen inequality therefore proves that the periodic corrector is zero and that the AP-E9 raw Cauchy–Born quartics are the homogenized densities. Their ratios are 4/3 and 3 on two gradients, so triangulation dependence survives homogenization. An eleven-case production scan then adds a physical translation quotient, dynamic centering, $a \downarrow 0.20833$, and $L \uparrow 7.5$. All cases are stationary, admissible, and have three-target degree one, but size, radial profile, and cross-triangulation gates fail. Consequently the complete Gamma-liminf, continuum background, same-action Hessian, and regulated determinant variation remain prohibited. The exact repair is structural: a compatible discrete de Rham/cup-product Skyrme term must replace the one-corner product before the continuum program can resume.

Contents

1 Declared stencil and proof obligations

Let $a > 0$ and write the forward differences on a cubic lattice as

$$D_i^a n(z) = \frac{n(z+e_i) - n(z)}{a}, \qquad i = 1, 2, 3. \tag{1.1}$$

The AP-E7 Skyrme density at the lower corner of a cube is

$$S_a(z) = \sum_{1 \le i < j \le 3} \left| D_i^a n(z) \wedge D_j^a n(z) \right|^2 . \tag{1.2}$$

It is sampled once per cube rather than assembled from the gradients of all tetrahedral interpolants. The associated algebraic forward current is

$$J_a(z) = \det\left[n(z), D_1^a n(z), D_2^a n(z), D_3^a n(z)\right]. \tag{1.3}$$

For genuine gradients, minors are null Lagrangians and div–curl arguments can give weak continuity under appropriate hypotheses [1]. Such an argument is not automatic for a finite difference product: the ordinary product rule contains shifts, and the unshifted expression in Eq. (1.2) is not a cochain cup product. Compatible discretizations instead preserve a differential complex and a bounded cochain projection [2]. AP-E10 tests the actual AP-E7 stencil, not an ideal compatible replacement.

Three statements must be separated:

(i) weak continuity of the two-minors in Eq. (1.2);

(ii) exclusion of a singular concentration measure for J_a;

(iii) identification of the weak limit of J_a with the geometric degree current of the normalized-affine map.

The first and third statements fail; a quantitative version of the second is true.

2 Exact failure of compensated compactness

2.1 A three-periodic two-minor defect

Work first on the unit torus with $a = (3m)^{-1}$. For lattice index $z = (z_1, z_2, z_3)$ put

$$\theta_z = \frac{2\pi}{3}(z_1 + 2z_2), \qquad n_a(z) = \left(\sqrt{1-\eta^2 a^2}, \eta a \cos\theta_z, \eta a \sin\theta_z, 0\right), \tag{2.1}$$

where $\eta > 0$ is fixed and $a < \eta^{-1}$. The sequence is exactly sphere-valued and $n_a \to e_0$ uniformly.

Lemma 2.1 (Forward phase identity)**.** *Let $f_z = \cos\theta_z$, $g_z = \sin\theta_z$, and suppose a forward step in direction i changes the phase by α_i. Then*

$$(\Delta_i f)(\Delta_j g) - (\Delta_i g)(\Delta_j f) = 4 \sin\frac{\alpha_i}{2} \sin\frac{\alpha_j}{2} \sin\frac{\alpha_j - \alpha_i}{2}. \tag{2.2}$$

Proof. Use $\Delta_i f = -2\sin(\alpha_i/2)\sin(\theta + \alpha_i/2)$ and $\Delta_i g = 2\sin(\alpha_i/2)\cos(\theta + \alpha_i/2)$, expand the determinant, and apply the sine subtraction identity. The result is independent of θ. □

Here $\alpha_1 = 2\pi/3$ and $\alpha_2 = 4\pi/3$. Therefore the target-component $(1, 2)$ entry of the spatial $(1, 2)$ Skyrme minor is exactly

$$(D_1^a n \wedge D_2^a n)_{12} = \frac{3\sqrt{3}}{2}\eta^2 \tag{2.3}$$

at every lattice site.

Theorem 2.2 (One-corner compensated compactness is false)**.** *There exists an $\mathbb{S}^3$-valued sequence converging uniformly, hence strongly in every finite L^p, to the vacuum, whose AP-E7 forward two-minors converge weakly to a nonzero constant although the continuum minor of the limit is zero. Its discrete Dirichlet and Skyrme energies remain uniformly bounded.*

Proof. The sequence (2.1) has $\|n_a - e_0\|_{L^\infty} = O(a)$. Its forward derivatives are $O(\eta)$ and its two-minors are $O(\eta^2)$; summation with cell volume a^3 therefore gives uniform Dirichlet and Skyrme bounds. Equation (2.3) contradicts weak continuity. Multiplication of the tangent components by a fixed $\chi \in C_c^\infty(\Omega)$ followed by exact normalization gives an exact-vacuum-boundary version whose defect tends to $(3\sqrt{3}/2)\eta^2\chi^2$ on the support of χ. □

For $\eta = 0.2$, the verifier obtains the same value 0.1039230484541327 at $a = 1/6, 1/12, 1/24, 1/48$, with maximum exact residual 5.6×10^{-17} while the uniform distance to the vacuum decreases to 4.17×10^{-3}.

2.2 A two-periodic diffuse topological-current defect

On an even lattice set

$$u_1 = (-1)^{z_1+z_2}, \qquad u_2 = (-1)^{z_2+z_3}, \qquad u_3 = (-1)^{z_3+z_1} \tag{2.4}$$

and define

$$n_a(z) = \left(\sqrt{1 - 3\eta^2 a^2}, \eta a u_1, \eta a u_2, \eta a u_3\right). \tag{2.5}$$

Again $n_a \to e_0$ uniformly and the image lies in the open northern hemisphere, so every geometric degree is zero. The tangent derivative matrix is

$$\eta \begin{pmatrix} -2u_1 & -2u_1 & 0 \\ 0 & -2u_2 & -2u_2 \\ -2u_3 & 0 & -2u_3 \end{pmatrix}. \tag{2.6}$$

Since $u_1 u_2 u_3 = 1$, its determinant is $-16\eta^3$. Thus

$$\boxed{J_a = -16\eta^3\sqrt{1 - 3\eta^2 a^2} \longrightarrow -16\eta^3,} \tag{2.7}$$

while the geometric degree and continuum current of the limit are zero.

For a fixed-boundary form, let

$$\chi(x) = \prod_{i=1}^{3} \sin^2(\pi x_i) \tag{2.8}$$

on $[0,1]^3$, replace the tangent part by $\eta a \chi u$, and normalize. Then the boundary is exactly the vacuum and

$$J_a \rightharpoonup -16\eta^3\chi^3 \mathrm{d}x, \qquad \int J_a \mathrm{d}x \longrightarrow -16\eta^3 \left(\frac{5}{16}\right)^3. \tag{2.9}$$

For $\eta = 0.2$ the limit is -0.00390625. At $a = 1/8$ and $1/48$ the computed integrals are respectively -0.00295055 and -0.00387605; the Dirichlet and Skyrme energies stay below 0.026 and 0.0024.

3 What the Skyrme bound still proves about concentration

Let A_a be the 4×3 matrix with columns $D_i^a n$, and let s_1, s_2, s_3 be its singular values. Pointwise,

$$|J_a| \le |\wedge^3 A_a| = s_1 s_2 s_3, \tag{3.1}$$

$$|\wedge^2 A_a|^2 = s_1^2 s_2^2 + s_1^2 s_3^2 + s_2^2 s_3^2. \tag{3.2}$$

Arithmetic–geometric mean gives

$$|\wedge^2 A_a|^2 \ge 3(s_1 s_2 s_3)^{4/3}. \tag{3.3}$$

Theorem 3.1 (No singular forward-current concentration)**.** *Every sequence with uniformly bounded AP-E7 Skyrme energy satisfies*

$$\boxed{\int_\Omega |J_a|^{4/3}\mathrm{d}x \le \frac{1}{3}\int_\Omega |\wedge^2 D^a n_a|^2 \mathrm{d}x.} \tag{3.4}$$

Hence J_a is weakly precompact in $L^{4/3}$ and cannot acquire an atomic or other singular-measure concentration defect.

Remark 3.2. Equation (3.4) excludes concentration but does not identify the weak limit. Equations (2.3) and (2.7) show that a diffuse oscillatory defect survives. Nor does Eq. (3.4) control the geometric normalized-affine current, because the AP-E7 action samples only one forward triad per cube. A theorem for geometric degree requires a compatible simplex-wise action and current. Recent two-dimensional charge-measure compactness results [3] demonstrate the needed architecture but do not transfer to this three-dimensional, noncompatible stencil.

4 Finite-R periodic cell problem

At the AP-E9 critical scale let $R_a = \gamma_a a^2/d_a^2 \to R \in (0,\infty)$. Around a fixed target point, a bounded microscopic tangent corrector gives a bond increment $a(Ar + D_r\varphi) + o(a)$. The logarithmic barrier expansion therefore yields

$$\mathcal{B}_a \longrightarrow \frac{R}{8}\int_\Omega Q_{\mathrm{hom}}(\nabla n)\mathrm{d}x. \tag{4.1}$$

Let $Y = (\mathbb{Z}/P\mathbb{Z})^3$ and let $\mathcal{B}_Y$ be the unique periodic bonds per cell, written as (p,r) from p to $p+r$. The exact convex cell problem is

$$Q_{\mathrm{hom}}(A) = \frac{1}{|Y|}\inf_{\varphi: Y\to\mathbb{R}^m}\sum_{(p,r)\in\mathcal{B}_Y}|Ar + \varphi(p+r) - \varphi(p)|^4. \tag{4.2}$$

The additive constant of φ is immaterial.

Definition 4.1 (Direction-wise cycle balance)**.** For each bond direction r, let $\mathcal{B}_{Y,r}$ be its directed periodic subgraph. It is cycle-balanced if every vertex has the same incoming and outgoing multiplicity.

Lemma 4.2 (Zero-corrector criterion)**.** *If every $\mathcal{B}_{Y,r}$ is cycle-balanced, then the zero corrector solves Eq.* (4.2) *exactly.*

Proof. Cycle balance gives

$$\sum_{(p,r)\in\mathcal{B}_{Y,r}}[\varphi(p+r) - \varphi(p)] = 0. \tag{4.3}$$

Because $v \mapsto |v|^4$ is convex, Jensen's inequality gives

$$\frac{1}{|\mathcal{B}_{Y,r}|}\sum_{(p,r)\in\mathcal{B}_{Y,r}}|Ar + D_r\varphi(p)|^4 \ge |Ar|^4. \tag{4.4}$$

Summing directions proves the lower bound, and $\varphi = 0$ attains it. □

The uniform six-tet graph has, per primitive cell, the seven directions

$$e_1, e_2, e_3, e_1+e_2, e_1+e_3, e_2+e_3, e_1+e_2+e_3. \tag{4.5}$$

The checkerboard five-tet graph has the three coordinate directions and both signs of every face diagonal with weight 1/2. Exact enumeration on the period-two cell gives nodewise in-degree equal to out-degree for every one of these direction graphs. Thus

$$Q_6^{\mathrm{hom}}(A) = \sum_i |Ae_i|^4 + \sum_{i<j} |A(e_i + e_j)|^4 + |A(e_1 + e_2 + e_3)|^4, \tag{4.6}$$

$$Q_5^{\mathrm{hom}}(A) = \sum_i |Ae_i|^4 + \frac{1}{2} \sum_{i<j} \big(|A(e_i + e_j)|^4 + |A(e_i - e_j)|^4\big). \tag{4.7}$$

The two checkerboard phases are lattice translates and have the same Q_5^{hom}. Random-start numerical corrector solves for four fixed and four random 3×3 gradients reproduce Eqs. (4.6)– (4.7) with maximum relative residual 9.1×10^{-15}.

Theorem 4.3 (Finite-R mesh dependence survives homogenization)**.** *The six- and five-tet homogenized densities are not scalar multiples. If $(Ae_1, Ae_2, Ae_3) = (v, 0, 0)$, then $Q_6^{\mathrm{hom}}/Q_5^{\mathrm{hom}} = 4/3$; if it is $(v, v, 0)$, the ratio is 3. No renormalization of the single coefficient R can equate them.*

This closes the AP-E9 cell problem, but negatively: a finite critical barrier is a mesh-dependent physical quartic. General discrete homogenization frameworks [4] justify the cell-formula architecture; here the cycle certificate makes the minimization explicit.

5 Translation quotient and dynamic centering

For a localized soliton define the nonnegative density, barycentre, and covariance

$$\rho = 1 - n_0, \qquad c[n] = \frac{\int x\rho \mathrm{d}x}{\int \rho \mathrm{d}x}, \qquad \Sigma[n] = \frac{\int (x-c)(x-c)^T \rho \mathrm{d}x}{\int \rho \mathrm{d}x}. \tag{5.1}$$

The translation quotient compares $\operatorname{tr}\Sigma$, its eigenvalues, and the normalized radial histogram about $c[n]$. It therefore identifies fields differing only by physical translation without adding an optimizer anchor. This distinction matters on growing boxes, for which ordinary strong-L^2 equicoercivity fails unless translations are quotiented or made tight.

The production driver also attempts a dynamic representative: after a stationary solve it shifts the field by $-c[n]$ using normalized linear interpolation, accepts the shift only if the logarithmic edge condition and all three geometric degree targets remain valid, and then re-relaxes the same action. Re-relaxation can return the field to a mesh-preferred Peierls position. Consequently the physical quotient, not a small raw barycentre, is the primary translation gate. Convergence of harmonic-map discretizations likewise requires declared triangulation and weight hypotheses rather than an informal refinement argument [5].

6 Production scan

The scan contains eleven independent unanchored solutions. The joint sequence is

N	L	a	total energy	centered RMS	gradient density
25	6.0	0.250000	76.672058	1.18660	3.04×10^{-7}
29	6.5	0.232143	77.679545	1.22405	1.64×10^{-6}
33	7.0	0.218750	78.552231	1.26582	5.34×10^{-7}
37	7.5	0.208333	78.987847	1.35422	1.70×10^{-6}

All eleven cases have direct tangent-plus-scalar gradient density below 2×10^{-6}, positive edge margin at least 0.1064, and three-target $B = (1, 1, 1)$. Production coverage and mechanical checks pass. The predeclared quotient diagnostics are

diagnostic	observed	threshold	pass
joint-tail energy spread	1.656%	3%	yes
joint-tail centered-RMS spread	9.612%	5%	no
joint maximum successive profile L^1	7.072%	5%	no
translated-start energy spread	0.00192%	1%	yes
translated-start centered-RMS spread	0.0602%	2%	yes
translated-start profile L^1	3.500%	3%	no
five-/six-tet energy spread	5.143%	3%	no
five-/six-tet centered-RMS spread	11.378%	5%	no

The maximum post-relaxation barycentre residual is $0.740a$, confirming Peierls locking and the need for the quotient. The cross-mesh discrepancy agrees in sign with the exact finite-R no-go and is not an optimizer failure.

7 Fail-closed decision and structural repair

Fail-closed gate 7.1 (Continuum and Hessian authorization)**.** The current AP-E7/AP-E8 lattice action cannot authorize a full Gamma-liminf, regulator-stable continuum background, Riemann Hessian, or regulated determinant variation because:

1. the one-corner two-minors are not weakly continuous;
2. the forward topological current has a nonzero diffuse defect at geometric degree zero;
3. finite-R five- and six-tet homogenized quartics are inequivalent;
4. the joint size/profile and cross-mesh numerical gates fail.

Thus

$$\texttt{forward\_minor\_compensated\_compactness} = \texttt{false}, \tag{7.1}$$
$$\texttt{finite\_R\_cell\_density\_computed} = \texttt{true}, \tag{7.2}$$
$$\texttt{finite\_R\_triangulation\_independence} = \texttt{false}, \tag{7.3}$$
$$\texttt{regulator\_stable\_background} = \texttt{false}, \tag{7.4}$$
$$\texttt{same\_action\_Hessian\_allowed} = \texttt{false}. \tag{7.5}$$

The next action should change the stencil rather than extend the same scan:

1. define dn as a primal one-cochain on the complete tetrahedral complex;
2. form the Skyrme two-form with a graded, shifted cup product satisfying a discrete Leibniz rule and $d^2 = 0$;
3. use a positive discrete Hodge star, averaged over the four body-diagonal six-tet meshes or calibrated by a cubic fourth-order design;
4. define the topological three-cochain from the same cup product and prove that its evaluation equals normalized-affine degree on admissible fields;
5. rerun the cell certificate and only then repeat the quotient continuum scan.

The FEEC principle is that stability requires a subcomplex and bounded cochain projection [2]; this is the appropriate first-principles repair. Historical lattice-Skyrme work already warns that a topology-preserving cutoff can become part of the model [6]. The finite GW/domain-wall kernel and actual $SO(3)$ mapping-torus mod-two index remain parallel lanes, and the degree-one Route-E portal remains last.

8 Reproducibility

Run from the repository root:

```
PYTHONPYCACHEPREFIX=/tmp/ap_e10_pycache python3 \
  route_f/code/scan_ap_e10_compactness_homogenization_centering.py
```

The deterministic result is written to

`route_f/output/ap_e10_compactness_homogenization_centering.json`

and its Markdown companion. The production card passes 10/10 exact, numerical, provenance, and fail-closed checks. A pass means that each claimed calculation was executed; it does not turn a deliberately false physics gate true.

References

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